[Figure 12 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 2** (the two-layer target: background $\mu^0 \approx 3\,\text{MPa}$, top layer $\mu^1 \approx 10\,\text{MPa}$), with **1 % noise** added to the measured surface displacements, using 3, 6, and 9 displacement datasets.

Explicit (left column): bilayer structure recovered with a relatively clean (though slightly wavy) interface. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 11.33]$ MPa; (b) 6 fields, SM $\in [1.00, 11.43]$ MPa; (c) 9 fields, SM $\in [1.00, 11.34]$ MPa.

Implicit (right column): bilayer structure visible but with significant artifacts, especially in panel (d) where a large green/blue patch intrudes into the red top layer from the left edge. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 14.60]$ MPa — a **46 % overshoot** above the nominal 10 MPa top layer, matching the paper's claim that the implicit method's average relative error "could exceed 40 %" under 1 % noise; (e) 6 fields, SM $\in [1.00, 10.60]$ MPa; (f) 9 fields, SM $\in [1.00, 10.71]$ MPa.

Takeaway: the noise amplifies the implicit method's difficulty with interface recovery dramatically in the 3-field case; 6 and 9 fields bring it back toward the nominal value but with visibly rougher interfaces than the explicit method.

Fig. 12. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 2 (1 % noise is introduced into the measured data, Unit: MPa).

[Figure 13 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 3** (the three-layer target: blue bottom layer $\mu^0 \approx 3\,\text{MPa}$, green middle layer $\mu^1 \approx 6\,\text{MPa}$, red top layer $\mu^2 \approx 10\,\text{MPa}$), with **1 % noise** added to the measured surface displacements, using 3, 6, and 9 displacement datasets.

Explicit (left column): three-layer structure recovered with wavy but distinct interfaces between the layers. Colorbar ranges: (a) 3 fields, SM $\in [1.01, 10.63]$ MPa; (b) 6 fields, SM $\in [1.00, 10.25]$ MPa; (c) 9 fields, SM $\in [1.00, 10.38]$ MPa.

Implicit (right column): three-layer structure noticeably noisier, with fuzzier interfaces and small artifact patches. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 11.78]$ MPa; (e) 6 fields, SM $\in [1.00, 11.58]$ MPa; (f) 9 fields, SM $\in [1.00, 9.96]$ MPa — notably, the 9-field implicit result **undershoots** the nominal 10 MPa target, unusual relative to most other panels in the paper which tend to overshoot.

Takeaway: as with Sample 2 under noise, the explicit method produces cleaner interfaces; but the quantitative colorbar-max values on Sample 3 are closer between explicit and implicit than on Sample 2 because the three-layer structure constrains both methods more tightly than the two-layer structure.

Fig. 13. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 3 (1 % noise is introduced into the measured data, Unit: MPa).

When 1 % noise was introduced, the explicit inverse method continued to perform admirably, producing high-quality reconstructed results (see Fig. 20). However, the performance of the implicit inverse method continued to be inferior to that of the explicit inverse method. The average relative error for the implicit inverse method could exceed 40 %, indicating substantial inaccuracies in the reconstructed results. On the other hand, the average relative error for the explicit inverse method remained relatively low, around 5 %, signifying its superior ability to handle the noise and produce more accurate reconstructions (see Table 3). The values of regularization factors used in each case from Fig. 19 and Fig. 20 are listed in Table 4.